

# Vinh Ly

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## Summary

Backend engineer with **17 years of experience** building distributed systems at YouTube, Google, and Microsoft. Focused on **backend services**, **cloud infrastructure**, and **data pipelines** that operate at enterprise scale. Currently working on YouTube backend infrastructure. Moving into solutions engineering to apply that depth directly to customer-facing technical work and enterprise sales cycles.

## Core Skills

**Languages:** Python, Go, Java, C#, TypeScript, Kotlin, SQL

**Cloud:** AWS (EC2, Lambda, S3, EKS), **GCP** (Cloud Run, BigQuery, Spanner), Azure (App Service, Table Storage)

**Databases:** PostgreSQL, MySQL, MongoDB, Redis, Google Spanner, Bigtable

**Data Processing:** **Apache Kafka**, Google Pub/Sub, Apache Beam, Dataflow, Flink

**Observability:** Datadog, Prometheus, Grafana, OpenTelemetry, Sentry

## Work Experience

### Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Backend infrastructure work on systems that serve **YouTube's content delivery at scale**. Built and maintained services in **Go** and **Python**, processing **over 1B+ events daily** across video metadata and content ingestion pipelines.
- Worked on distributed data systems handling large-scale video catalog operations for the platform.

### Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- Platform team lead (7 engineers) during the live commerce platform's early growth phase. Designed the catalog and order management services using **Kotlin**, **Apache Kafka**, and **PostgreSQL**, supporting thousands of concurrent streamers and millions of daily transactions during peak sales events.
- Rebuilt the product data ingestion pipeline from batch-oriented to a streaming architecture. **Cut seller-facing data latency from several hours to under 2 minutes**, enabling near-real-time inventory display.
- Introduced Kubernetes cost optimizations and spot instance strategies. **Monthly cloud spend dropped by around \$20K** without affecting reliability.
- Transitioning focus toward solutions-oriented engineering work and customer-facing technical engagements.

### Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Owned backend services, data pipelines, and API infrastructure for **Google Shopping**, serving millions of daily users. Worked on product catalog ingestion, serving layers, and search ranking integration points.
- Led a 6-engineer team redesigning the product data ingestion pipeline. **Pipeline latency fell from several hours to under a minute**; the system processed millions of products per day and was later adopted as the standard for Shopping Merchant Center integrations.
- Reviewed pull requests daily and coached three junior developers on API design and SQL performance tuning. One eventually moved into a tech lead role on a sibling product team.

### Software Engineer | Google | Hybrid | 02/2011 - 02/2015

- Full-stack backend work on **Google Shopping**, primarily in **Python** and **Java**. Built data pipeline components and internal tooling used by the product and merchant-facing teams.

- Shipped several features for the product data indexing system that improved coverage and freshness for merchant-uploaded catalogs.

## **Software Design Engineer II | Microsoft | Remote | 10/2007 - 12/2011**

- Full-stack engineer on the **Office Online** platform team. Built real-time document collaboration features in **C#** and **SignalR**, handling the backend infrastructure that supported millions of monthly active users.
- Contributed to the document storage and retrieval backend services powering Office Web Apps.

## **Education**

**BS Computer Science | University of Texas at Dallas | 2003 - 2007**